Empirical Alignment of LLM-Derived NGAP FSMs Using an Open-Source 5G Testbed for Specification-Grounded Protocol Fuzzing*

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Abstract

This study presents a pilot evaluation of the alignment between LLM-derived NG Application Protocol (NGAP) finite state machines (FSMs) and real execution traces from an open-source 5G testbed. The FSMs were extracted from NGAP procedure text in 3GPP TS 38.413 and filtered into procedure-level core transitions. The evaluation covers NG Setup and UE Context Release using PCAP and log evidence. Across 16 in-case FSM transition-case pairs, 14 were exercised in the actual traces, and all 14 exercised transitions aligned with the observed behavior, yielding 14/14 alignment among exercised transitions. The overall trace coverage was 14/16, or 87.5%. Non-triggered and non-evaluable transitions were reported separately to clarify the interpretation scope of the results. These results should be interpreted as evidence of feasibility for the evaluated procedures, rather than as a complete validation of LLM-derived NGAP FSM correctness. They suggest that specification-derived FSM transitions can be grounded in implementation traces and used as a structured basis for future fuzzing feedback analysis.

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1 Introduction

NG Application Protocol (NGAP) is the control-plane protocol used on the N2 interface between an NG-RAN node and the AMF in 5G systems [1]. For protocol fuzzing, a finite state machine (FSM) extracted from the specification can guide state-aware mutation, but only if the extracted transitions are consistent with actual implementation behavior. This study evaluates whether LLM-derived NGAP FSM transitions can be aligned with real execution traces from an open-source 5G testbed [2], [3]. Unlike network-protocol fuzzer evaluations such as AFLNet [4], this study focuses on a narrower prerequisite question: whether specification-derived FSM transitions can be compared with PCAP and log evidence before being used in a closed-loop fuzzing workflow.

Figure 1 shows the broader closed-loop NGAP fuzzing architecture considered in this work. The architecture consists of four modules: FSM generation, mutation strategy generation, fuzzing execution, and self-reflection. This paper does not add a validation module to the architecture; instead, it performs an offline FSM-to-trace alignment analysis of the FSM JSON produced by the FSM Generation Module before it is used as a transition reference for future fuzzing feedback analysis.

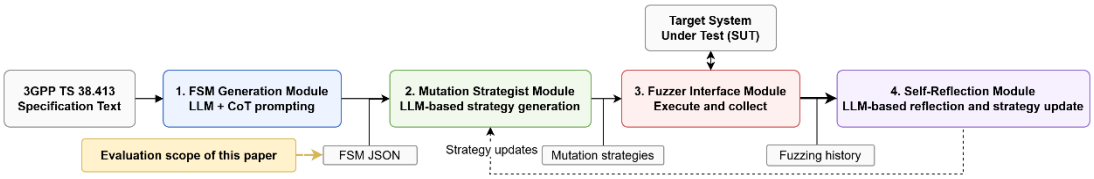


Figure 1: Proposed closed-loop NGAP fuzzing architecture and the evaluation scope of this study.

2 Methods and Results

Gemini 3.1 Pro Preview was used with a fixed prompt template to extract FSM transitions from NGAP procedure text in 3GPP TS 38.413 [1]. Each extracted transition was represented using source state, trigger event, target state, and guard condition. A core transition was defined as a procedure-level state change triggered by an NGAP message or guard condition. During manual filtering, transitions were retained only when they represented procedure-level state progression; IE parsing/checking steps and self-loops without state progression were removed, and semantically duplicate transitions with the same source, trigger, guard, and target were merged. After filtering, NG Setup contained 12 core transitions and UE Context Release contained 10 core transitions.

The testbed used Open5GS v2.7.7 and UERANSIM v3.2.8 in Docker [2], [3]. This pilot study focuses on high-level transition patterns of NG Setup and UE Context Release rather than release-specific IE-level differences. For each case, SCTP-filtered PCAPs and AMF/gNB/UE logs were collected. Transition-case pairs were labeled as Matched, Guard-matched, Not triggered, Mismatch, N/A, or N/E. Matched and Guard-matched were counted as alignment success. Not triggered transitions were included in the in-case FSM-pair denominator when they were relevant to the evaluated case but not exercised in the analyzed trace. In contrast, transitions marked as N/A or N/E were reported descriptively and were not included in the in-case FSM-pair denominator.

This study distinguishes alignment among exercised transitions from trace coverage. Alignment among exercised transitions is calculated as Aligned / Exercised in trace. This metric excludes transitions that were not triggered or not evaluable in the tested workflow. In contrast, trace coverage is calculated as Exercised in trace / In-case FSM pairs. This distinction avoids conflating FSM-to-trace

Table 1: FSM-to-trace alignment results

alignment with the execution scope of the testbed. Table 1 summarizes the FSM-to-trace alignment and trace coverage results across the evaluated cases.

Procedure	Case	In-case FSM pairs	Exercised in trace	Aligned	Not triggered	Aligned / Exercised	Exercised / In-case
NG Setup	Normal	5	5	5	0	100.0%	100.0%
NG Setup	PLMN mismatch	7	5	5	2	100.0%	71.4%
UE Context Release	AMF-initiated	4	4	4	0	100.0%	100.0%
Total	-	16	14	14	2	100.0%	87.5%

As shown in Table 1, 14 of the 16 in-case FSM transition-case pairs were exercised in the actual traces, and all 14 exercised transitions aligned with the observed traces. Therefore, the alignment among exercised transitions was 14/14, or 100.0%, while the overall trace coverage was 14/16, or 87.5%. This result should not be interpreted as evidence that all in-case transitions were exercised or that the complete NGAP FSM was validated.

In the normal NG Setup case, all five case-relevant transitions were exercised and aligned with the FSM. In the PLMN mismatch case, NGSetupFailure and the TimeToWait IE were observed, while two retry-related transitions were treated as Not triggered because they were not included in the analyzed trace. For UE Context Release, the AMF-initiated case exercised all four case-relevant transitions, whereas six RAN-initiated paths were marked as N/E because they could not be exercised in the tested UERANSIM workflow.

3 Discussion

This pilot study shows that LLM-derived NGAP FSM transitions can be compared with real execution traces for the evaluated procedures. By separating alignment among exercised transitions from trace coverage, the study avoids conflating FSM-to-trace alignment with the execution scope of the testbed. The result is not a claim of complete or general correctness of LLM-derived NGAP FSMs. Instead, it indicates that specification-derived transitions can serve as practical trace-level references for the evaluated NGAP procedures and as a basis for future fuzzing feedback analysis.

The experiment also revealed execution-scope limitations. In the PLMN mismatch case, retry-related transitions after NGSetupFailure were not exercised in the analyzed trace, and RAN-initiated UE Context Release could not be exercised naturally in the tested UERANSIM workflow. These cases show that FSM-to-trace alignment is useful not only for reporting aligned transitions, but also for identifying non-triggered transitions and testbed limitations.

This study has several limitations. It evaluates only two NGAP procedures, uses a single Open5GS/UERANSIM testbed, and relies on manually filtered LLM-derived transitions that were not independently validated by multiple reviewers. This study does not evaluate fuzzing effectiveness, vulnerability discovery, or mutation guidance improvement. Therefore, the result should be interpreted as a pilot alignment study for the evaluated procedures rather than a complete validation of NGAP FSM correctness or the general correctness of LLM-derived NGAP FSMs. Future work will extend this evaluation to more NGAP procedures and integrate the alignment process into a closed-loop fuzzing framework.

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